

Cybersecurity Project

CloudBased AD Setup and User Management

Objective

The objective of this project is to give learners practical, hands-on experience with setting up and managing a cloud-based Active Directory environment—an essential skill for SOC analysts and Windows security practitioners. By completing the lab, learners will gain a solid understanding of core AD concepts such as domains, forests, and user management. They will provision virtual machines in the cloud, install and configure Active Directory Domain Services, and create a functioning domain from scratch. Learners will also join a client computer to the domain, create and manage user accounts, and perform basic administration tasks that mirror real-world enterprise operations. This project prepares learners to monitor domain activities, recognize authentication events, and understand how centralized directory services support security operations within modern organizations.

Required Tools / Prerequisites

To complete this project, learners will need:

- **Cloud provider free tier** (Azure, AWS, or Google Cloud Platform). Azure offers a 12-month free tier including Windows VMs.
- **Two Windows Server VMs** (can be trial versions, e.g., Windows Server 2022 Evaluation).
- **Remote Desktop (RDP)** to access the VMs.
- Basic knowledge of Windows Server and networking.

Common Azure Free Account Blockers (Quick Tips)

During Azure free account creation and initial VM deployment, learners may encounter the following common blockers. These quick tips help prevent setup delays.

Card Verification Issues

Issue: Card is declined during signup.

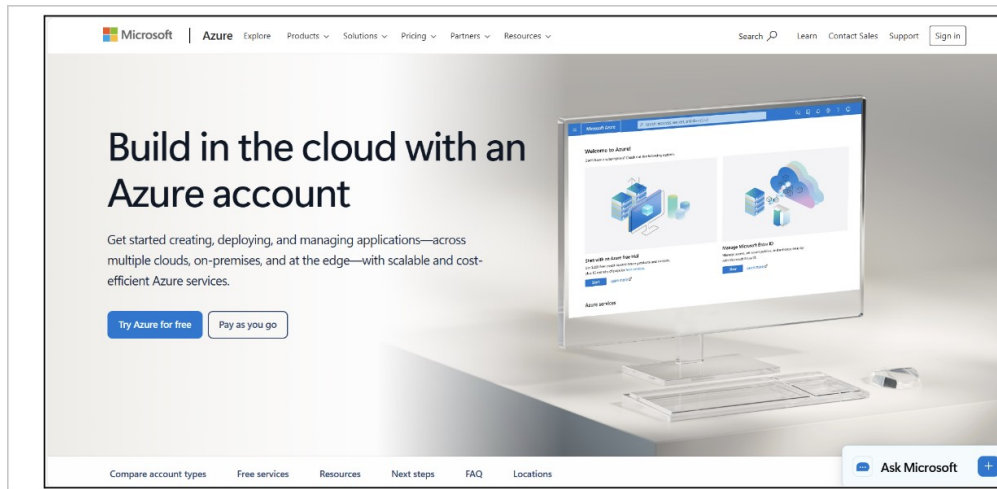
Workaround: Use a supported card, either credit or debit (Visa, Mastercard, American Express, JCB).

Instructions

Before creating your Windows VMs, you must set up your Azure environment. Azure's free tier gives you enough credits and resources for this project.

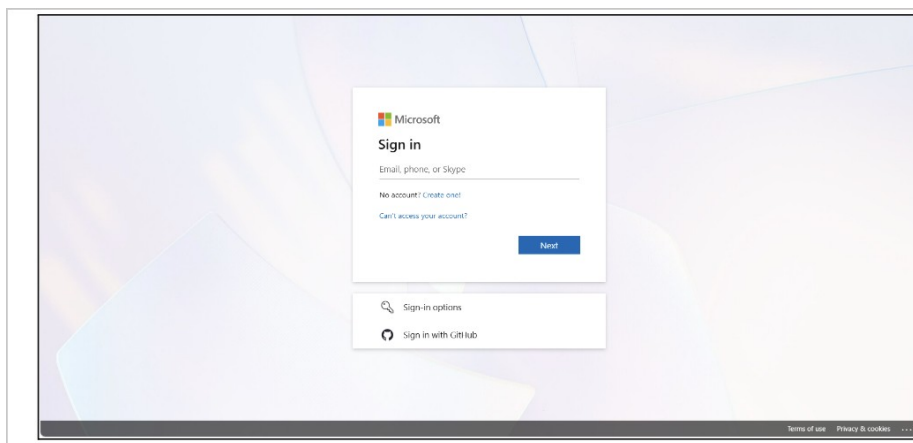
Step A: Create a Microsoft Azure Free Account

1. Go to: <https://azure.microsoft.com/free>
2. Click Try Azure for Free



azure.microsoft.com/free — "Try Azure for Free"

3. Sign in using a Microsoft account
 - If you don't have one, click Create one!



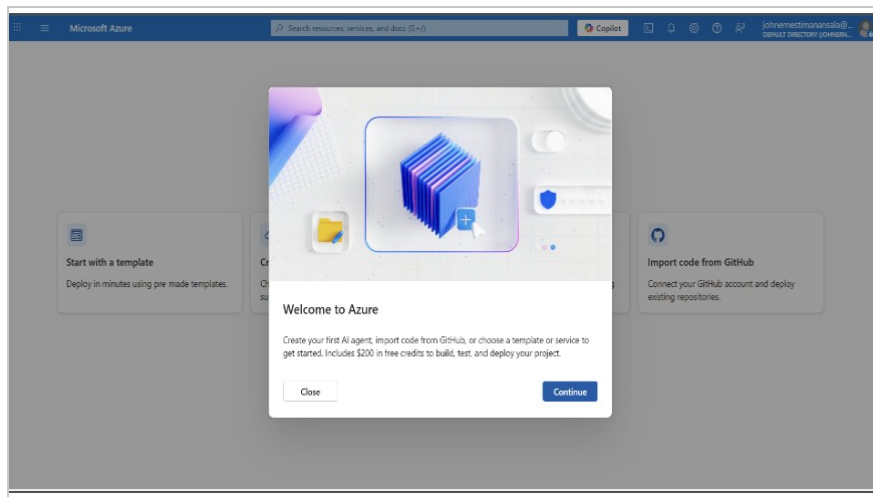
Microsoft sign-in page

4. Provide:
 - Use of Azure Account: For Personal Use
 - Country
 - Name
 - Phone number (SMS verification required)

- Address Information
5. Enter payment information
 - Only for identity verification.
 - Only accepts the following cards (Credit or Debit): Visa, Mastercard, American Express, JCB.
 - **You will NOT be charged** unless you manually upgrade.
 6. Accept terms → Click **Sign Up**

Step B: Access the Azure Portal

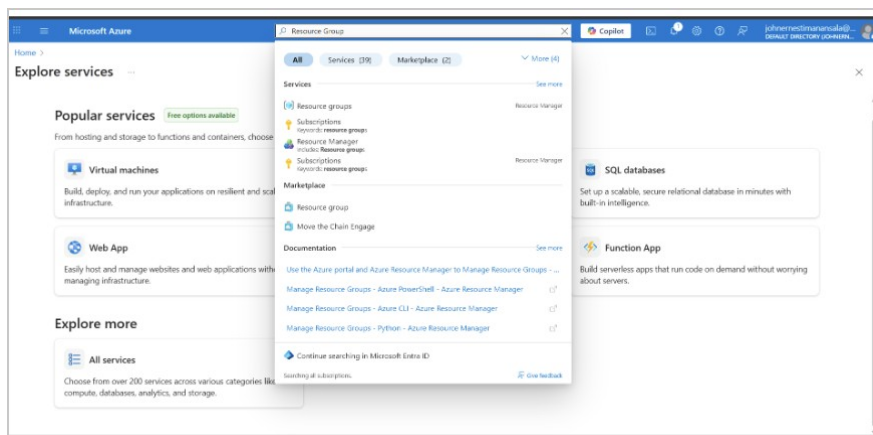
1. Go to <https://portal.azure.com>
2. Sign in to your account
3. You should now see the Azure dashboard



Azure dashboard — “Welcome to Azure”

Step C: Create a Resource Group

1. Search for **Resource Groups**



Searching for Resource groups

2. Click **Create**
3. Configure:
 - **Subscription:** Free Trial

- **Resource Group Name:** AD - Lab
- **Region:** your nearest region (e.g., Southeast Asia)

4. Click **Review + Create** → **Create**

Create a resource group — Basics

Step D: Create a Virtual Network (VNet)

Both VMs must be on the same private network.

1. Search for **Virtual Network**
2. Click **Create**
3. Configure:
 - **Resource Group:** AD - Lab
 - **Name:** AD - VNet
 - **Region:** same as Resource Group
4. Leave the default Address Space and Subnet
5. Click **Review + Create** → **Create**

Create virtual network — Basics

With the Resource Group and VNet ready, you can now proceed to Step 1, where you will deploy the Domain Controller and client VM.

Step 1: Provision Cloud Virtual Machines

Create two Windows Server VMs in your account:

VM1 – Domain Controller (DC01)

Purpose: Hosts Active Directory Domain Services (AD DS), DNS, and serves as the primary domain controller for the lab environment.

Before you deploy: Your Image is currently set to “Windows Server 2022 Datacenter Server Core – x64 Gen2.” Server Core has no desktop GUI, so you will not be able to use Server Manager, the Add Roles and Features wizard, or the notifications-flag promotion steps in Step 2 the way this guide describes. Change the Image to “Windows Server 2022 Datacenter – x64 Gen2” (Desktop Experience — the option without “Server Core” in the name) before you click Review + create.

My Configuration (as deployed):

- **Virtual machine name:** DC01
- **Region:** (Asia Pacific) New Zealand North
- **Availability options:** Availability zone → Self-selected zone → Zone 1
 - Optional for a single-VM lab — fine to leave at the default if you’d rather skip zone selection.
- **Security type:** Trusted launch virtual machines (default — keep this; it enables Secure Boot and vTPM on the Gen2 image).
- **Image:** Windows Server 2022 Datacenter **Server Core** – x64 Gen2 — **change to the Desktop Experience edition (see note above)**
- **VM architecture:** x64
- **Size:** Standard_B2als_v2 – 2 vCPUs, 4 GiB memory (~\$36.43/month; price varies by region/currency)
- **Administrator account:** Username **azureuser** (or your own choice), with a strong password
- **Public inbound ports:** Allow selected ports → RDP (3389)
 - Azure warns this opens RDP to all IP addresses. For a lab this is common, but consider restricting the source to your own IP under the Networking tab → network security group rules.
- **Licensing (Azure Hybrid Benefit):** “Use an existing Windows Server license” is checked — leave this **unchecked** unless you actually hold an eligible on-prem/CSP Windows Server license.

Create a virtual machine ...

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Instance details

Virtual machine name * ⓘ

Region * ⓘ [Deploy to an Azure Extended Zone](#)

Availability options ⓘ

Zone options * ⓘ

☒ Self-selected zone
Select a specific zone for your VM

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

Availability zone * ⓘ

Security type ⓘ [Configure security features](#)

Image * ⓘ [See all images](#) | [Configure VM generation](#)

VM architecture * ⓘ

☐ Arm64

☒ x64

Size * ⓘ [See all sizes](#) | [Customize cores](#)

Your Create a virtual machine — Basics (Instance details)

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Run with Azure Spot discount ⓘ ☐

Administrator account

Username * ⓘ

Password * ⓘ

Confirm password * ⓘ

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☐ None

☒ Allow selected ports

Select inbound ports *

ⓘ **This will allow all IP addresses to access your virtual machine.** This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Licensing

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#) ⓘ

Use an existing Windows Server license ⓘ ☒

To apply this Azure Hybrid Benefit, you must confirm that you have an eligible Windows Server license with Software

Your Create a virtual machine — Administrator account, inbound ports, licensing

Still to configure (Networking tab — not shown in your screenshots):

- Network: **AD-VNet**

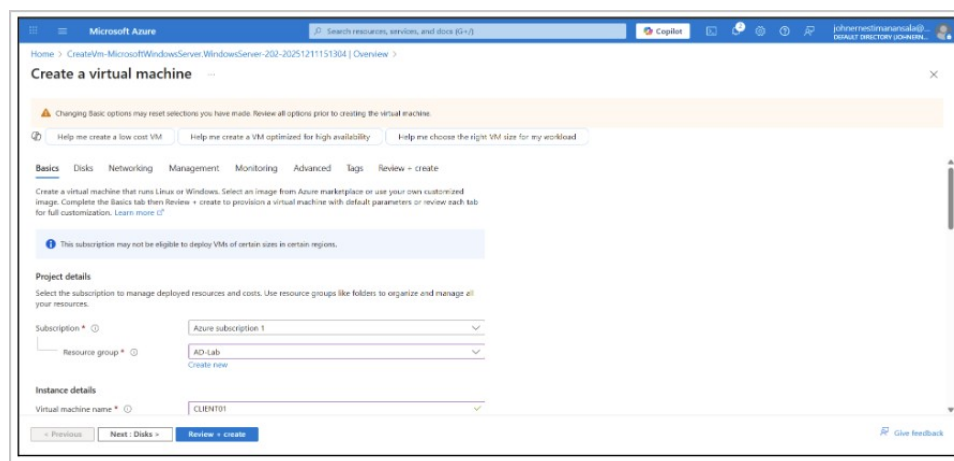
- **Subnet:** default
- **Public IP:** Optional (recommended only for RDP access; use NSG restrictions)
- **Private IP:** Static (recommended for domain controllers)
- **Notes:** Ensure time sync and DNS settings point back to the DC itself.

VM2 - Client Machine (CLIENT01)

Purpose: Acts as a domain-joined workstation used to test authentication, Group Policies, and user account management.

Configuration:

- **OS:** Windows 10, Windows 11, or Windows Server 2022
- **Size:** B2als_v2
 - vCPUs: 2
 - RAM: 4 GB
 - Temp Storage: 4 GiB
 - Network Bandwidth: 3750 Mbps
 - Cost Estimate: ~\$41.17/month
 - Reason for Choosing: Matches DC performance, ensures smooth RDP usage, and is stable for client operations.
- **Network:** AD-VNet
- **Subnet:** default
- **Private IP:** Dynamic is acceptable
- **Inbound Ports:** RDP (3389)
- **DNS Settings:** Must point to DC01's private IP before joining the domain
- **Notes:** Keep OS patched before domain join for secure configuration.



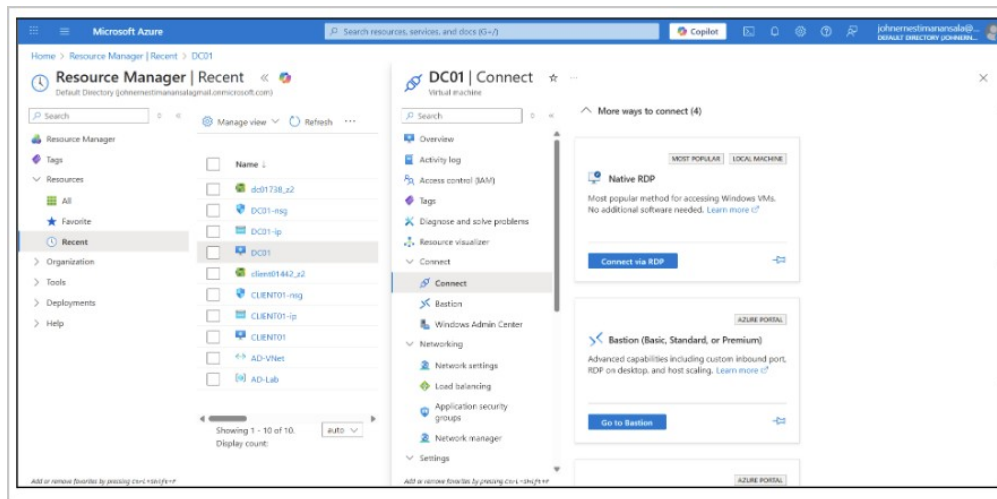
Create a virtual machine — CLIENT01

Both VMs must be in the same virtual network (AD-VNet) and the same region. This allows:

- Domain join
- DNS resolution
- Group Policy communication
- Realistic enterprise LAN simulation

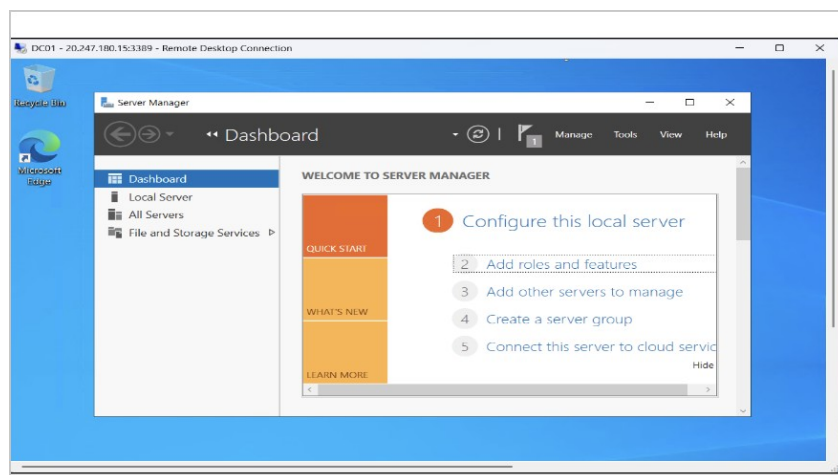
Step 2: Configure Domain Controller

1. Log in to VM1 (Domain Controller) via RDP.
 - Go to **Virtual Machines** → select **DC01** → click **Connect**.
 - Download the RDP file and open it.



Azure portal — DC01 Connect (Native RDP)

- Use the credentials you set when creating the VM.
- The Remote Desktop session will open the Windows Server desktop.



DC01 desktop — Server Manager Dashboard

2. Open **Server Manager** (usually starts automatically at login; if not, click the Start Menu → Server Manager).
 - Click **Add Roles and Features**.

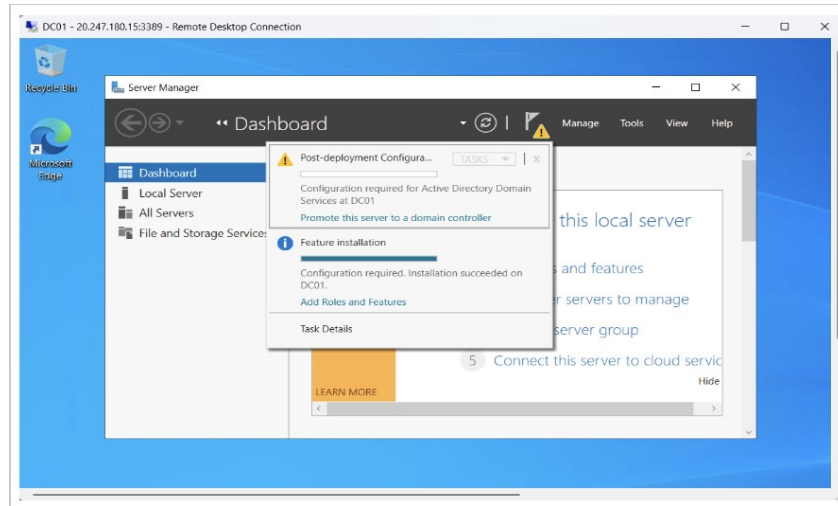
In the Add Roles and Features Wizard:

- **Installation Type:** Select Role-based or feature-based installation → Next.
- **Server Selection:** Select DC01 → Next.
- **Server Roles:** Check Active Directory Domain Services → a pop-up will ask to add features → click Add Features → Next.
 - **Features:** Leave default → Next.

- **AD DS:** Read description → Next.
- **Confirmation:** Check “Restart the destination server automatically if required” → click Install.

Note: The installation might take a few minutes.

3. After installation, click the **Notifications flag** in Server Manager → **Promote this server to a domain controller.**



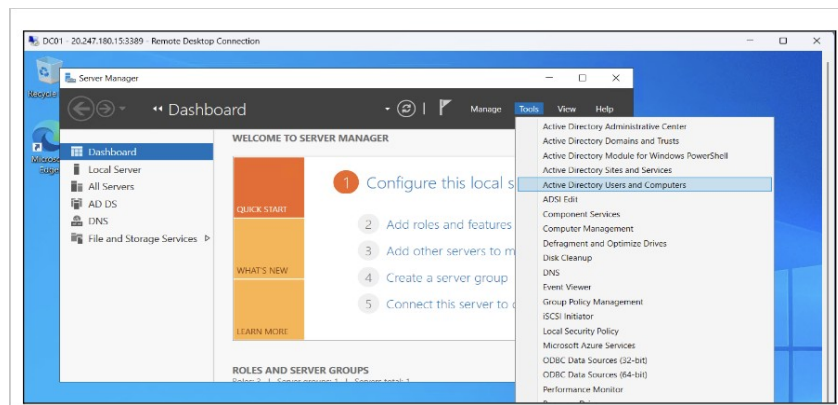
Server Manager notification — Promote this server to a domain controller

- **Deployment Configuration:**
 - Choose **Add a new forest** → Root domain name: **corp.local** → Next.
- **Domain Controller Options:**
 - Forest and Domain Functional Levels: Default (Windows Server 2022).
 - Check DNS Server → Next.
 - Set DSRM password (Directory Services Restore Mode). Remember this password; it's used for recovery → Next.
- **Additional Options:** Leave default → Next.
- **Paths:** Leave default (C:\Windows\NTDS, etc.) → Next.
- **Review Options:** Confirm → Next.
- Click **Install** → The server will automatically reboot after promotion.
- After reboot, DC01 is now the Domain Controller for the new forest **corp.local**.

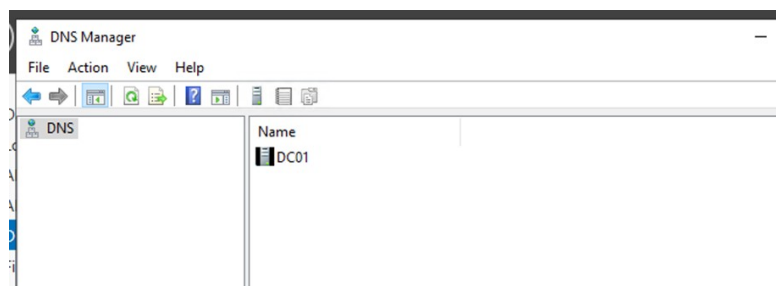
Note: During the domain setup, a domain administrator account named “adminuser” is created. This account has Domain Admin privileges and is used throughout the lab for administrative tasks.

4. Verify AD DS and DNS

- Log back into DC01.
- Open **Server Manager** → **Tools** → **Active Directory Users and Computers**.
- You should see your domain corp.local listed.
- Open **DNS** from Server Manager → Tools → **DNS** to check that your domain zones are created automatically.



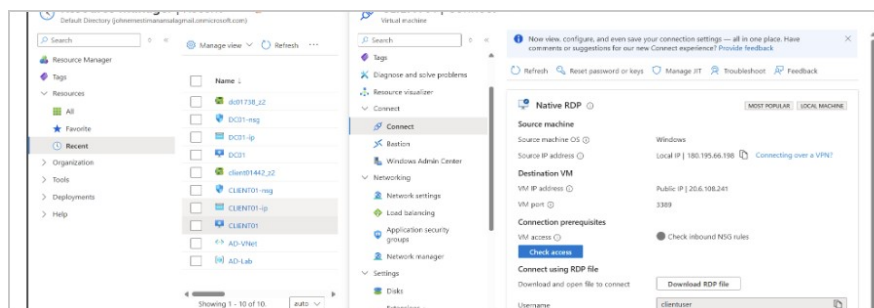
Server Manager → Tools → Active Directory Users and Computers



Tools → DNS

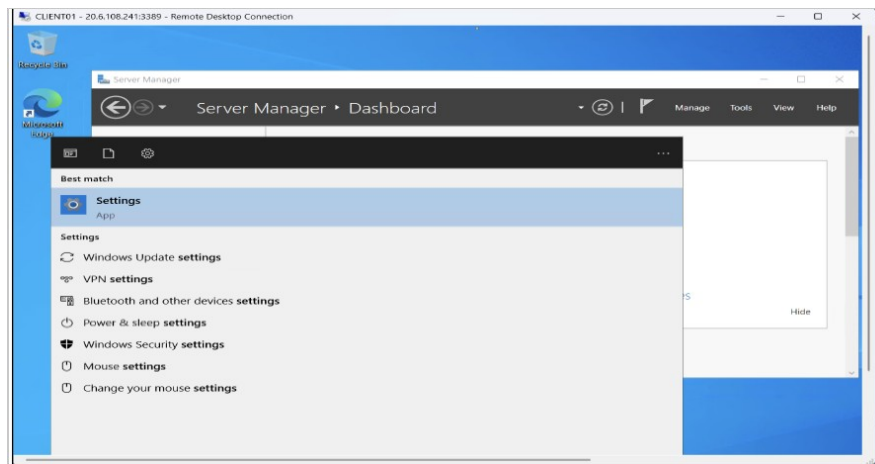
Step 3: Join Client VM to the Domain

1. Log in to VM2 (Client machine).
 - Do the same with VM1 and download the RDP.
 - Open RDP to **CLIENT01** using the credentials set during VM creation.
 - **Ensure CLIENT01's DNS Server is set to DC01's private IP:**

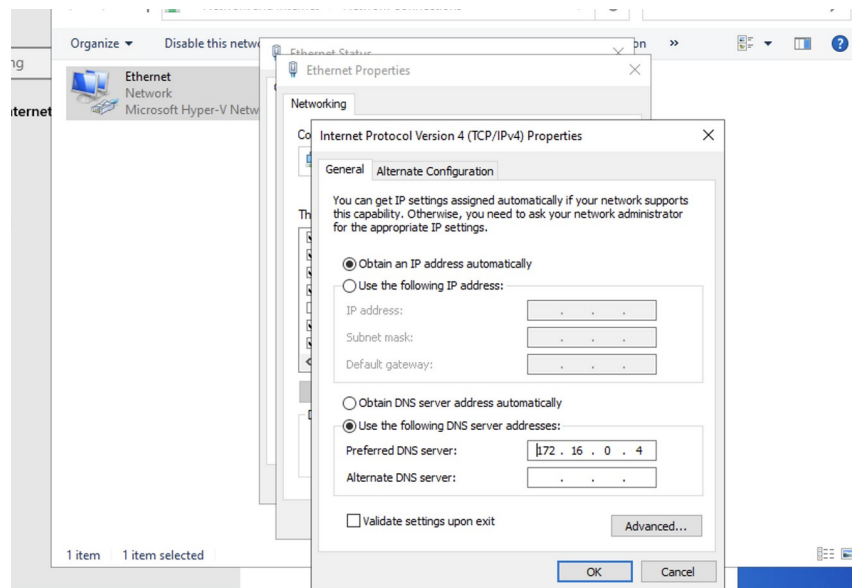


Azure portal — CLIENT01 Connect

- Open **Settings → Network & Internet → Ethernet → Change adapter options → Right-click your adapter → Properties → IPv4 → Properties → Preferred DNS server → Enter DC01 private IP → OK.**



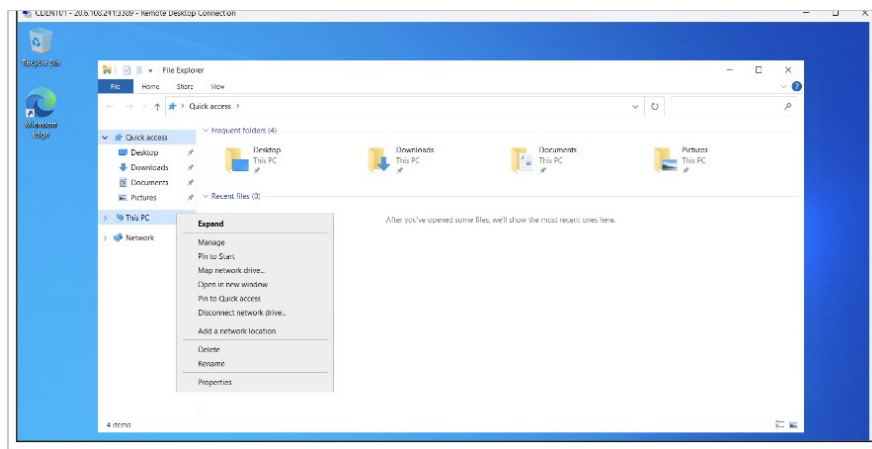
CLIENT01 — searching for Settings



DC01 private IP: 172.16.0.4

2. Open System Properties:

- Right-click **This PC** → **Properties** → **Advanced system settings** → **Computer Name** tab → **Change...**



CLIENT01 — This PC → Properties

- **Select Domain**
 - Enter the domain: **corp.local**
 - Click **OK**.
- Enter **Domain Admin credentials** (adminuser) from DC01.
 - A pop-up will ask for credentials → type the **Domain Admin** username and password from VM1 (DC01)
 - Click **OK**
- Restart CLIENT01 to apply domain membership.
 - A message will appear saying the computer must restart to join the domain → click **Restart Now**

After reboot, the workstation is now domain-joined.

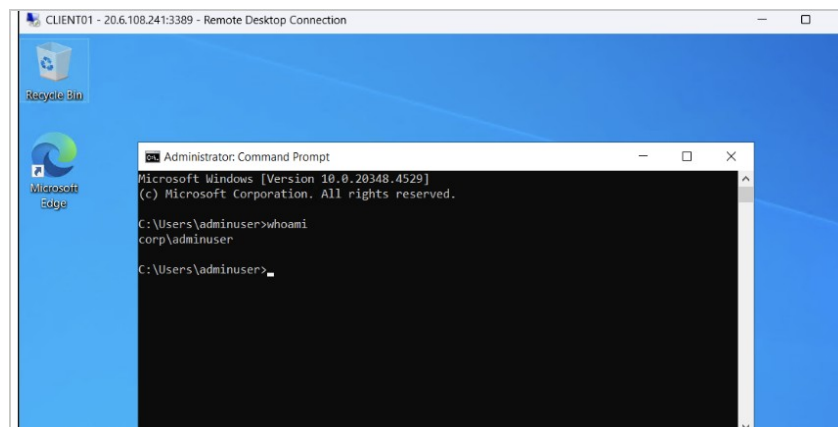
3. Verify the Domain Membership

Log in with domain credentials:

- On the login screen, type:
 - **Username:** **corp\adminuser**
 - **Password:** The same one from DC01

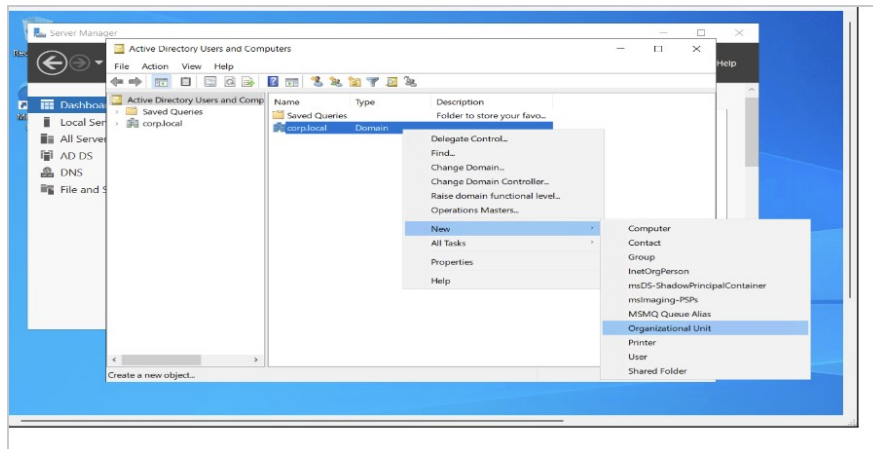
Check the domain account in the Command Prompt:

- **Type:** **whoami**
- **Output:** **corp\adminuser**



Step 4: Create Users and Organizational Units (OUs)

1. On VM1 (DC01), open **Active Directory Users and Computers**.
 - Server Manager → Tools → **Active Directory Users and Computers (ADUC)**.
2. Right-click your domain → New → **Organizational Unit**
 - Example: **TestUsers**



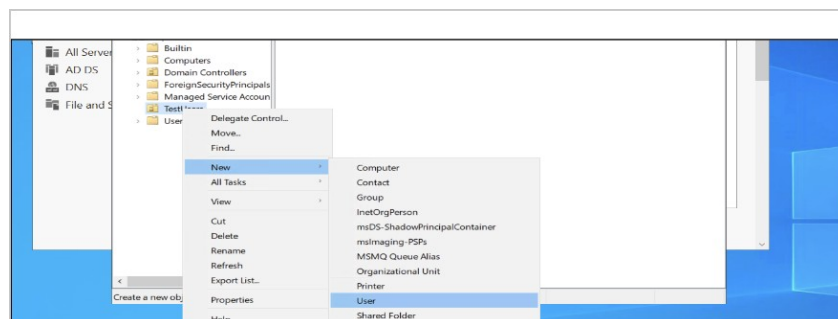
ADUC — New → Organizational Unit

3. Right-click **TestUsers OU** → New → **User**.

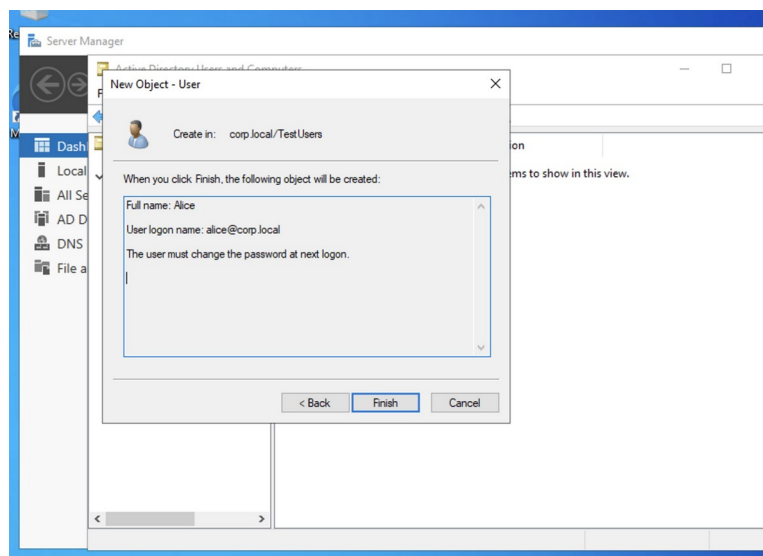
Inside the OU, create users:

- **Alice**
 - First name: Alice → Last name: N/A → User logon name: alice → Next
 - Set password → Require password change at next login → Next → Finish
 - Repeat for Bob and Charlie.
- **Bob**
- **Charlie**

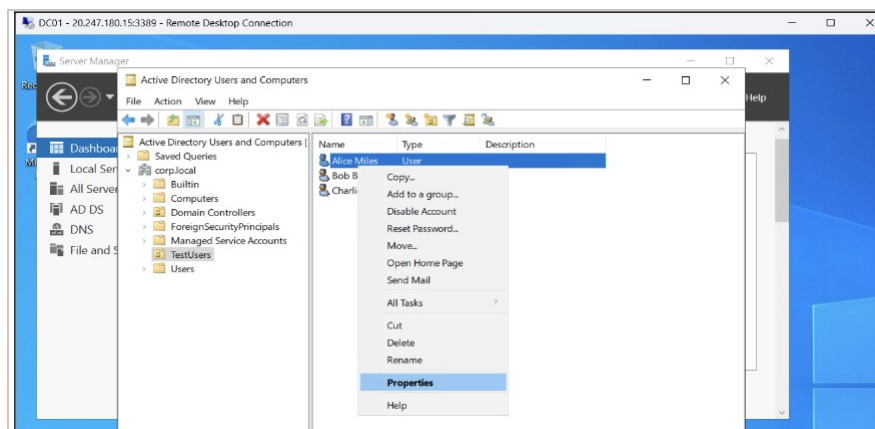
Ensure all users are added to the **Domain Users** group (default).



ADUC — New → User



4. In ADUC, right-click a user → **Properties** → **Member Of** → Add to custom groups if needed.



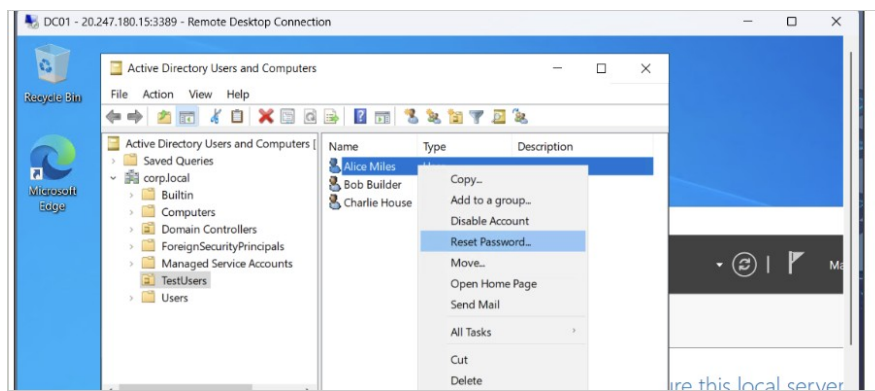
ADUC — user context menu → Properties

This simulates user provisioning and organizational structure, both monitored by SOC analysts during investigations.

Step 5: Reset Password & Add User to “Remote Desktop Users” Group

1. On DC01: Open **Active Directory Users and Computers**
 - Server Manager → Tools → **Active Directory Users and Computers**
 - Go to: **corp.local** → **TestUsers** → **alice**
 - Right-click **Alice** → **Reset Password**
 - Set a **new password**
 - **UNCHECK**: User must change password at next logon
 - Click **OK**

This fixes the issue with changing the password at next logon.

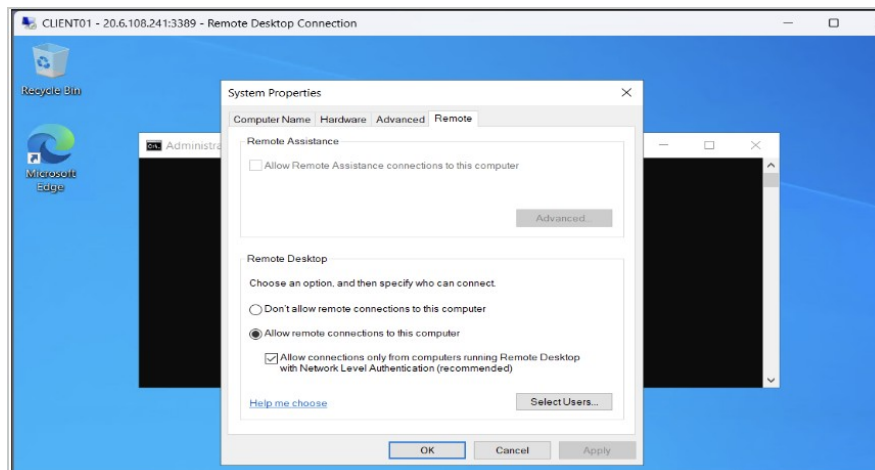


ADUC — Reset Password

2. On **CLIENT01** (the machine you're RDPing into):

- Log in using an **Administrator** account (adminuser)
 - Example: **corp\adminuser**
- Open **System Properties**
 - Press Win + R
 - Type: **sysdm.cpl**
 - Press Enter
- Go to the **Remote** tab
 - Click **Select Users...**
 - Click **Add**
 - Enter: **corp\alice**
 - Click **Check Names** → **OK** → **OK**

This adds Alice to the **Remote Desktop Users** local group.



System Properties — Remote tab

Step 6: Test User Accounts

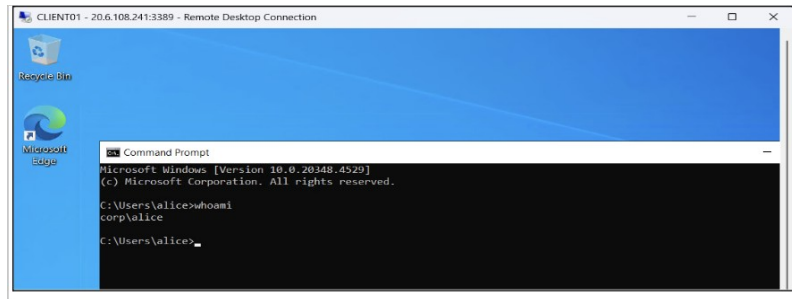
Ensure users can log in and communicate with domain resources.

1. On **CLIENT01** (VM2), log out of Administrator → log in with a new domain account, e.g., **corp\alice**.

- Enter the username and password created in Step 4.

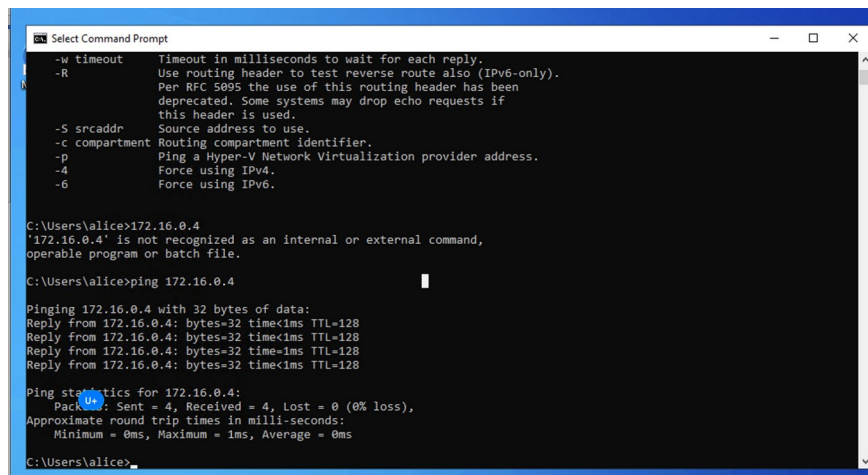
2. Verify Domain Access

- Open **Command Prompt** → run: **whoami**



CLIENT01 — whoami as corp\alice

- **Output: corp\alice**
- **Test network access:**
 - Press **Windows + E** to open **File Explorer**
 - Click the **address bar** at the top
 - Type: **\\DC01** → Press Enter
 - Access shared folders (if any), but you may see a blank window (no shares yet)
 - Open Command Prompt again (or reuse the same one)
 - Type: **ping <DC01IP Address>** → Should reply.



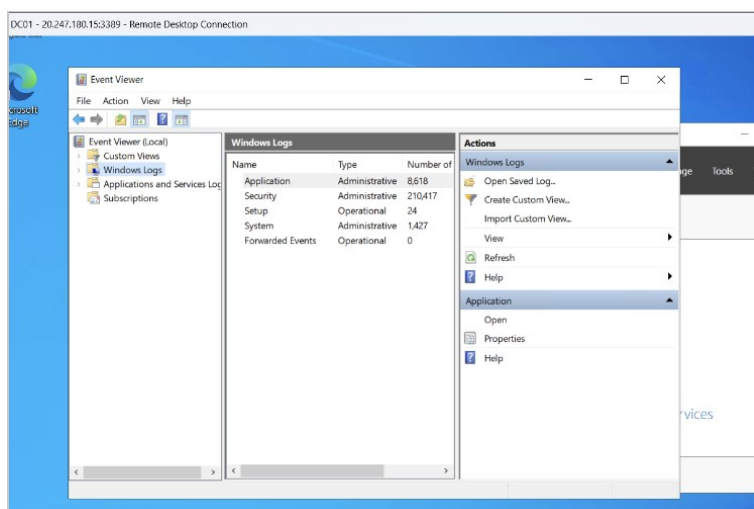
CLIENT01 — ping DC01 succeeds

Step 7: Monitoring

1. Review Security Logs on the Domain Controller

- Log in to **DC01** as a domain administrator (adminuser): **corp\adminuser**
- Open **Event Viewer**:
 - Click **Start Menu**
 - Search for **Event Viewer**
 - Open the application
- **Navigate to:**

- **Event Viewer → Windows Logs → Security**



DC01 — Event Viewer → Windows Logs

This log contains **authentication events, account management activity, and policy changes**, which are critical for SOC investigations.

2. Identify Key Active Directory Security Events

Use **Filter Current Log...** (right panel) and search for the following Event IDs:

Authentication Events

Event ID	Description	SOC Relevance
4624	Successful logon	Confirms valid user authentication
4625	Failed logon	Detects brute-force or password guessing
4648	Logon using explicit credentials	Indicates lateral movement attempts

Account Management Events

Event ID	Description	SOC Relevance
4720	User account created	Tracks user provisioning
4722	User account enabled	Detects reactivated accounts
4723 / 4724	Password change/reset	Monitors credential changes
4725	User account disabled	Tracks access revocation
4726	User account deleted	Identifies account removal

Domain & Group Changes

Event ID	Description	SOC Relevance
4732	User added to a security group	Privilege escalation monitoring
4733	User removed from a security group	Access change tracking

Event ID	Description	SOC Relevance
4735	Security group modified	Detects group tampering

Important Note: Event ID Variations

Some Windows Security Event IDs may vary depending on:

- Windows Server version (e.g., 2016, 2019, 2022)
- Client OS version (Windows 10 vs Windows 11)
- Domain Functional Level
- Audit Policy configuration (Advanced Audit Policy settings)
- Whether the event occurred on the **Domain Controller** or the **client machine**

For example:

- Authentication events (e.g., **4624 / 4625**) may appear on both the **client** and **DC**, but with different **Logon Types**.
- Group membership changes (e.g., **4732**) may be logged on the **Domain Controller** even if the action was initiated from a client.
- Password-related events may display as 4723 or 4724, depending on whether the change was initiated by the user or the administrator.

3. Correlate Events With Lab Actions

Map events to actions performed earlier:

Lab Action	Expected Event
Password reset	4724
Account logs in	4624
Scheduled Task was Updated	4702
Special privileges assigned to the new logon.	4672
Group Membership Information	4627
A new process has been created	4688

Note: Some events may not appear unless the corresponding action is performed. These examples illustrate what SOC analysts commonly monitor in enterprise environments.

Correlation Tip:

If an expected Event ID is not found, verify:

- The action was performed successfully
- You are reviewing logs on the correct system (DC01 vs CLIENT01)
- Security auditing is enabled for the relevant category

Analysts commonly correlate multiple related events across systems to confirm activity.

4. Export Logs for Analysis

- In **Event Viewer** → **Security**

- **Right-click Security**
- **Click Save All Events As...**
- **Save as: DC01-SecurityLogs.evtx**

Analysts often export logs to SIEM platforms such as Splunk, Sentinel, or ELK for centralized analysis and alerting.

Optional Azure Security Tools (Free-Tier Friendly)

In addition to the native Windows Event Viewer, Azure provides built-in security and monitoring tools that can be explored within the free-tier subscription. These tools help learners understand how cloud-native security controls integrate with Active Directory environments.

Explore additional Azure security features within free-tier credits.

1. Microsoft Defender for Cloud

Purpose: Cloud security posture management and basic threat protection.

What can be explored:

- View security recommendations for virtual machines
- Identify exposed management ports (e.g., RDP 3389)
- Review baseline security posture and compliance suggestions

SOC Relevance:

Introduces how security teams identify misconfigurations and prioritize risks in cloud-hosted Windows environments. The free tier includes foundational security posture assessments. Advanced workload protection is optional and not required for this lab.

2. Azure Network Security Groups (NSG)

Purpose: Network-level traffic filtering for cloud resources.

What can be explored:

- Restrict RDP access to a specific IP range
- Understand inbound vs outbound security rules
- Simulate basic perimeter hardening for domain controllers

SOC Relevance:

Demonstrates how access control and network segmentation reduce attack surfaces and support incident response. NSGs are included at no additional cost and are fully usable within free-tier subscriptions.

Summary

By completing this project, learners will:

- Provision and configure a cloud-based Active Directory environment using Microsoft Azure, including virtual networks, resource groups, and Windows virtual machines.
- Install and promote a Windows Server to a Domain Controller, creating a new Active Directory forest and configuring DNS for enterprise identity management.

- Join a client machine to the domain and verify domain membership, authentication, and network communication.
- Create and manage Organizational Units (OUs) and user accounts, simulating real-world user provisioning and access management workflows.
- Troubleshoot common Active Directory issues such as DNS misconfigurations, domain join failures, and user login problems.
- Develop foundational SOC-relevant skills by understanding how centralized identity, authentication events, and domain activity are used to support security monitoring and incident investigations.